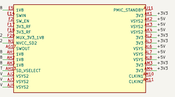

USB
CAN
CAN_RX
CAN_EN
PCIE_RF
MUX_INT1_1V8
WAKE_INT1_1V8
WAKE_INT2
LED
LED
SD_VSELECT
SD_VSELECT
VDD12
VDD12
VDD12
PWRC_STANDBY
3V3
3V3
VSYS
VSYS
VSYS
VSYS
3V3
3V3
VSYS
VSYS
3V3
3V3
CLKOUT
CLKOUT
+VBB_AA
+VBB_AB
+VBB_AC
SD_VSELECT
+5V_AA
+5V_AB
+5V_AC
+3V3
+5V
+5V
+5V
+3V3
+3V3
+5V
+5V
+3V3
+3V3
WAKE_SD2
WAKE_SD2

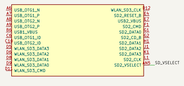
ENET1_MDC
ENET1_MDIO
ENET1_RXD3
ENET1_RXD2
ENET1_RXD1
ENET1_RXD0
ENET1_RX_CTL
ENET1_RXC
ENET1_TXD3
ENET1_TXD2
ENET1_TXD1
ENET1_TXD0
ENET1_TX_CTL
ENET1_TXC

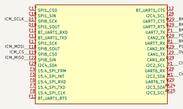
ENET2_TX_CTL
ENET2_TXC
ENET2_TXD3
ENET2_RXD3
ENET2_RXD2
ENET2_TXD2
ENET2_TXD0
ENET2_TX_CTL
ENET2_TXD1
ENET2_RXD0
ENET2_RXC
ENET2_RXD1

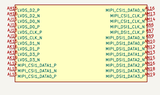
PCM_WCLK
SAI3_WCLK
SAI3_RXD
SAI3_TXD
SAI3_TXC
PCM_CLK
PCM_OUT
PCM_IN

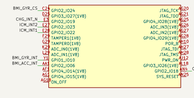
BT_ANT_EXT
WAKE2_JTAG_TDI
WAKE2_JTAG_TDO
WAKE2_JTAG_TCK
WL_DEV_WAKE
BT_DEV_WAKE
WAKE2_JTAG_TRST
WL_HOST_WAKE

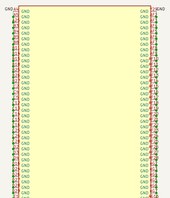
J1
+3V3
ICM_SCLK
ICM_MOSI
ICM_CS
IMU_INT2
BMI_SCLK
BMI_MOSI
BMI_SDO
BMI_SDI
BMI_GYR_CS
BMI_ACC_INT
GND
GND
ICM_MISO
GND
IMU_INT2
GND
BMI_SDO
BMI_ACC_CS
GND
BMI_GYR_INT
IMU_SOCKET

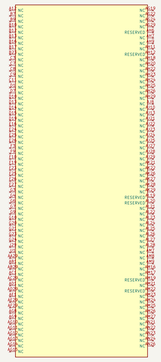

USB_OTG1_N
USB_OTG1_P
USB_OTG2_N
USB_OTG2_P
USB_OTG_ID
WLAN_SD3_DAT3
WLAN_SD3_DAT2
WLAN_SD3_DAT1
WLAN_SD3_DAT0
WLAN_SD3_CMD
WLAN_SD3_CLK
SD2_RESET_B
USB3_RBIAS
SD2_CMD
SD2_CLK
SD2_DAT3
SD2_DAT2
SD2_DAT1
SD2_DAT0
SD2_VSELECT

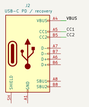

SPI1_CS0
SPI1_SIN
SPI1_SCK
SPI1_SOUT
SPI0_CS0
SPI0_SCK
I2C1_SDA
I3S_A_SPI_FRM
I3S_A_SPI_INT
I3S_A_SPI_RXD
I3S_A_SPI_TXD
I3S_A_SPI_CLK
BT_UART1_RTS
BT_UART1_CTS
I2C1_SCL
UART1_CTS
UART1_RTS
UART_WL_WLP_RXD
CAN2_TX
CAN2_RX
CAN1_TX
CAN1_RX
I2C2_SCL
I2C2_SDA
UART0_RX
UART0_TX
UART0_CTS
UART0_RTS
I2C3_SCL
I2C3_SDA
ICM_SCLK
ICM_MOSI
ICM_CS
ICM_MISO
BT_UART1_CTS
BMI_MOSI
BMI_SCLK
BMI_ACC_CS
FG_FAULT
BMI_SDO
BMI_SDI
CHG_INT_N
I2C_CHG_SCL
I2C_CHG_SDA

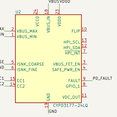

WIFI_CSI1_DATA0_N
WIFI_CSI1_CLK_N
WIFI_CSI1_CLK_P
WIFI_DSI1_CLK_N
WIFI_DSI1_CLK_P
WIFI_DSI1_DATA0_N
WIFI_DSI1_DATA0_P
WIFI_DSI1_DATA1_N
WIFI_DSI1_DATA1_P
WIFI_DSI1_DATA2_N
WIFI_DSI1_DATA2_P
WIFI_DSI1_DATA3_N
WIFI_DSI1_DATA3_P
WIFI_CSI1_DATA1_N
WIFI_CSI1_DATA1_P


GPIO2_IO24
GPIO3_IO27(1V8)
GPIO2_IO19
GPIO2_IO20
GPIO2_IO21
GPIO2_IO22
TAMPER1(1V8)
TAMPER2(1V8)
ADC_IN0(1V8)
ADC_IN1(1V8)
GPIO1_IO10
GPIO1_IO11
GPIO4_IO19(1V8)
GPIO4_IO15(1V8)
ON_OFF
JTAG_MOD
JTAG_TCK
GPIO4_IO28(1V8)
ADC_IN3(1V8)
ADC_IN2(1V8)
GPIO4_IO18(1V8)
PCAL_A
JTAG_TDI
JTAG_TMS
PWR_ON
GPIO3_IO26(1V8)
GPIO2_IO23
SYS_RESET
BMI_GYR_CS
BMI_GYR_INT
BMI_ACC_INT
CHG_WT_A
ICM_INT2
ICM_INT1
CHG_OE_N


GND
GND


NC
RESERVED
RESERVED
RESERVED
RESERVED


J2
USB-C PD // recovery
VBUS
CC1
CC2
SBU1
SBU2
VBUS
SHIELD
GND


U2
VBUS_MAX
VBUS_MIN
ISNK_COARSE
GND
CC1
CC2
ISNK_COARSE
ISNK_FINE
VBUS_FET_EN
SAFE_PWR_EN
FAULT
GPIO_1
VCC_DIG
CYPD3177-24LQI
VBUS
VDD
VBUS_MIN
VBUS_MAX
ISNK_COARSE
HPI_SCL
HPI_SDA
INT
FAULT_N
PD_FAULT

U3
BQ25798
VBUS
VBUS
VAC1
ACDRV1
VAC2
ACDRV2
SW1
SW2
BTST1
BTST2
SW2
SW1
CHG_SCL
CHG_SDA
CHG_INT_N
CHG_CE_N
STAT
REGN
SYS
SYS
BAT
BAT
PMID
PMID
CHG_PMID
CHG_BTST1
CHG_SW1
CHG_SW2
CHG_BTST2
CHG_STAT_N
REGN
VSYS_CHG
VBAT
CHG_PROG
BAT_TS
SCL
SDA
INT
CE
QON
VAC2
VAC1
PGND
PGND
SYS
SYS
BATP
BATP
GND

U4
AW37503ADNR
VSYS_CHG
VSYS_CHG
VCC_BUCK
IN
IN
EN
VCC
BOOT
SW
FB
PG
GND
BOOT_5V
SW_5V
FB_5V
PG_5V
GND

U5
TPS62913RWUR
VIN
VIN
SW
SW
PGND
PGND
AGND
VOS
FB
EN
PG
VOUT
VOUT
VOUT
VOUT
GND
+5V_RAW
+5V_RAW
OVP_UV
OVP_OV
OVP_SENSE
OVP_FB
OVP_FLT_N
OVP_EN
OVP_SENSE
OVP_FLT_N
OVP_FB
OVP_UV
OVP_OV
OVP_THR
OVP_EN
OVP_THR
OVP_SENSE
LTC4365MS#PBF

U8
GND
GND

NC
RESERVED
RESERVED
RESERVED
RESERVED

VDD
VBUS_MIN
VDD
VBUS_MAX
VDD
ISNK_COARSE
VBUS_MIN
VBUS_MAX
ISNK_COARSE
GND
GND
GND

+3V3
I2C_SDA
+3V3
I2C_SCL
+3V3
CHG_INT_N
REGN
CHG_CE_N
CHG_PROG
GND

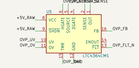
BOOT_5V
C3
2.2UF
SW_5V
C5
2.2UF
+5V_RAW
VCC_BUCK
C4
4.7UF
GND
VSYS_CHG
C1
47UF
GND
+3V3
C2
4.7UF
PG_5V
GND

+5V_RAW
FB_5V
R1
GND
FB_5V

+5V_RAW
C7
10uF
GND
BAT_TS
R2
GND
VBAT
BAT_TS
GND
2S LiFePO4 BAR + BMS

+5V_RAW
OVP_HGATE
Q1
NMOS pass - TBD
OVP_SENSE

OVP_SENSE
R3
+5V


OVP_FB
OVP_FLT_N
OVP_THR
R4
R5
R6
R7
R8
R9
R10
R11
R12
R13
R14
R15
R16
R17
R18
R19
R20
R21
R22
R23
R24
R25
R26
R27
R28
R29
R30
R31
R32
R33
R34
R35
R36
R37
R38
R39
R40
R41
R42
R43